COMP3208 Social Computing (UoSM) 2025/2026
Coursework II

Module Code:	COMP3208		
Module Title:	Social Computing Techniques		
Module Leader:	Asst.Prof. Dr Fairuz Safwan Mahad		
Assessment Type:	Individual	**Weightage:**	25%
Submission Due Date:	**4th May 2026, 17:00 (Malaysia Time)**		
Method of Submission:	Blackboard		
GenAI Tier Specification	**Tier 1: No GenAI use [applies All - Entire work]**		

This assessment relates to the following Module Learning Outcomes:

B. Subject Specific Intellectual and Research Skills	B1. Use recommender technologies such as item-based and user-based collaborative filtering techniques
D. Digital Skills	D1. Exhibit social computing experiments and analyse the results using a scientific approach using of recommender technologies

Academic Integrity

"In this entire coursework, students are **prohibited** from **using any GenAI tools** for their **assessed work**. This includes entering any part of the assignment or your assessed work to GenAI, whether by pasting/typing, uploading files, or describing content directly or through plugins. Basic tools that assist spelling and grammar, translation and calculation without generating new content or ideas, can be used unless specified otherwise by the assessment setter. GenAI may be used to explain lecture slides and notes to enhance understanding of a relevant topic areas. Students are not required to complete a GenAI Declaration Form."

For more information on GenAI refer here

The aim of this coursework is to promote learning. You are encouraged to discuss concepts with peers and refer to external resources. However, all source code must be your own original work. Use of AI-generated code (e.g., ChatGPT, Bard) will result in penalties according to university policies. Ensure your submissions reflect your own understanding and effort.

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Coursework II (Matrix Factorization)

This Coursework is worth **25%** of the total marks for this module. The deadline is **4th May 2026, 17:00 (Malaysia Time)**, hand-in via the blackboard site. The work is expected to take **10 to 30 hours**, can vary based on your familiarity with python and problem-solving skills.

Introduction:
This coursework, like real life, is open-ended. We will guide you through some things you need to know, helping in the support periods, and you will work out what you need to learn and how to learn it. The coursework will require you to familiarise yourselves with the basics of recommendation system to build your own recommendation system.

Coursework Overview:
The aim of the coursework is to develop and evaluate realistic recommender system algorithms. Individual submissions are expected for this coursework (i.e. not group work). Source code must be a single .py file. No other formats are acceptable. Strictly refrain from using third party libraries, except for numpy. For third party library, ONLY numpy can be installed as it is essential in manipulating arrays. This coursework is intended to evaluate your ability to understand and develop recommender system algorithms from scratch, not your ability to use powerful third-party libraries (scikit-learn, etc.) to do the task efficiently without a deep understanding.

You are also expected to create a report (**1000-1300 words**) detailing various components that helps readers understand the whole algorithm and implementation process. Components may encompass, but are not restricted to, **Methodology**, **Analysis**, **Processes**, **Dataset**, etc. The documentation format is left to your discretion, allowing you the freedom to determine the most suitable components for providing a clear understanding of your work. The primary objective of the documentation is to showcase a profound comprehension of the underlying processes of the recommendation system and the ability to implement it from the ground up. Documentation is also critical in the industry as it is required to archive the entire process for future reference. It is also required to explain the methodology, analysis and to justify the results to the team, client and shareholders. This coursework is designed to simulate the experience of working in the industry.

Rules:
- Individual work.
- Keep your code clean, complete, and easy to understand.
- A zip file, “cw2.zip” is expected to be submitted.
- Late submissions will be penalised at 10% per working day.
- No work can be accepted after feedback has been given.
- Submitted work must be the student’s ORIGINAL WORK
- Any kind of plagiarism, such as copying an existing work from the internet, or copying from/to other
- students will be penalized under the regulation of the University of Southampton.
- Use of chatGPT, Bard, CoPilot, Blackbox, or any other AI generated content is strictly prohibited and if found, will be penalised

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Matrix Factorization

Use the matrix factorization recommender system algorithm to train and predict ratings for a large (20M) set of items. You may need to use a database to handle the large data.

Resources:

You are provided with the following:

1.	cw2_template.py	python code template
2.	test_20M.csv	testing dataset (20% from 20M) (Format: user_id (int), item_id (int), timestamp (int))
3.	train_20M.csv	training dataset (80% from 20M) (Format: user_id (int), item_id (int), rating (float), timestamp (int))
4.	result_sample.csv	Example of result (Format: user_id (int), item_id (int), rating_prediction (float), timestamp (int))
6.	env_win.yml	conda environment file for windows OS

A Python template file is provided to get you started with. The "train_20M.csv" file includes user_id, item_id, rating, and timestamp, while the "test_20M.csv" is missing the rating column. Use the "train_20M.csv" file to train using the matrix factorization recommender system algorithm and predict the ratings in the "test_20M.csv" file. Please refer to the result_sample.csv file for an example of the expected output. The output of your source code must be a .csv file in the following format:

Output format:

File Type	.csv			
columns	user_id (int)	item_id (int)	rating_prediction (float)	timestamp (int)

Deliverables:

Files to submit	cw2.zip | → 1. cw2_submission.py | → 2. cw2_report.docx

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Please ensure to rename the cw2_template.py to cw2_submission.py before submitting. Also ensure to submit a cw2_report.docx to demonstrate that you have a deep understanding of the entire process of your work.

Code Execution and Virtual Environment:

You are provided with one anaconda environment .yml file for Windows OS which can be used for coursework II. Please feel free to generate an environment based on the relevant file corresponding to your operating system. You may create your own environment from scratch or even use another virtual environment you are comfortable with. However, it is essential to ensure that your source code runs error-free in the provided environment file because evaluation will be done using the provided environment file. In the case that you are creating an environment from scratch or if you are using another environment, please ensure that your environment is installed with python version 3.9.18 and numpy version 1.26.4. To maintain fairness, refrain strictly from using packages and libraries that are not listed in the provided environment file. Should you encounter any problems running or installing the provided environment files, please reach out to the lecturer for assistance.

If using conda/pyenv: conda env create -f env_win.yml

Benchmark Mean Absolute Error (MAE) scores for tasks:

A benchmark MAE score will be computed using our predicted ratings and the ground truth ratings. Your predicted ratings will be used to compute your MAE score against the ground truth ratings. The ground truth rating remains undisclosed to you. Your MAE score will then be compared against the benchmark MAE score for evaluation. Beating the benchmark MAE score will result in a 100% mark for each task. The following provides an idea on the performance of previous year students. However, these values may change as a different dataset will be given. Please take note that the lower the MAE score the better.

Benchmark MAE scores for 2025:

Cw2 average MAE score 0.65, top MAE score 0.60

Notes and Restrictions:

Make sure that you provide a prediction for each of the rows in the test set. Failure to provide a complete set will result in missing predictions, which will be set to zero by default and cause a higher MAE rate than would otherwise be achieved.

Given the size of the large dataset, we recommend that you use a database. A simple database such as SQLite suffices.

Marking Guidelines & Feedback

This coursework accounts for 25% of the total marks for this module and will be assessed according to the attached rubrics. Marks and written feedback will be provided within 3 weeks of the due date. Please note that the marks are provisional and subject to change after moderation.

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Marking Guidelines

Grading Criteria	Wtg	Excellent 5	Very Good 4	Good 3	Average 2	Fair 1	Poor 0	Mark (≈Wtg x Score)
Coursework 2	**8.0**	MAE score is better than the benchmark score	MAE score is as good or similar to the benchmark score	MAE score is lower than benchmark score and the difference between the benchmark score is within 20-39%	MAE score is lower than benchmark score and the difference between the benchmark score is within 40-50%	MAE score is lower than benchmark score and the difference between the benchmark score is above 50%	Code does not produce a .csv file	**/40**
Documentation	**2.0**	Provides a well-organized and clearly documented report with comprehensive explanations.	Presents a well-organized report with adequate documentation.	Presents a report with some organization but lacking in documentation.	Presents a disorganized report with minimal documentation.	Fails to present a coherent report or document the process.	No presentation or documentation.	**/10**
Total								**/50**

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Any work submitted after the deadline's time will be subject to the standard University late penalties unless an extension has been granted, in writing by the Senior Tutor, in advance of the deadline. Details on the University's late penalties can be found here:

- https://www.southampton.ac.uk/~assets/doc/quality-handbook/Late%20Submission.pdf

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